

Nisarg Trivedi

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Education

Dhirubhai Ambani University

B.Tech, Information and Communication Technology.

Gandhinagar, Gujarat

2024-2028

Bellevue High School

GPA: 3.87

Bellevue, Washington

June, 2024

Experience

Speech Research Lab, DAU

Gandhinagar, Gujarat

Research Assistant

May 2025 – Present

- Conducted data collection and preprocessing to build robust speech corpora for research and model training.
- Developing APIs enabling seamless integration of speech synthesis and classification modules
- Designed and trained speech language models for Dysarthria severity classification, improving clinical applicability of speech technology
- Researched and implemented GAN-based speech synthesis approaches, enhancing naturalness and intelligibility of generated speech.
- Working on audio LLMs for multilingual speech enhancement and low-resource language settings.

Google Developer Groups on Campus, DAU

Gandhinagar, Gujarat

Core Member

November 2024 – Present

- Organized flagship event *Devolution*, featuring expert speakers across diverse technology domains, with 600+ participants
- Coordinated event logistics, speaker outreach, and participant engagement as part of the core organizing team
- Conducted technical sessions on Android development using MIT App Inventor for beginner developers
- Delivered introductory workshop on Generative AI fundamentals tailored for incoming freshmen.

Projects

- **DiaTongue** - Built an AI-powered diabetes screening system using MobileNetV2 and FastAPI, deployed via a Telegram bot for real-time tongue image analysis with 86% accuracy.
- **PRI-Prescription-Risk-Insight** - Built a Random Forest model to classify opioid prescriptions using Data by Washington State Government for High/Low Risk using demographic and prescription data, achieving 87% accuracy.
- **Talk2DAU** - Developed a Streamlit-based conversational AI chatbot for university resources using LangChain, Groq LLM, and FAISS, enabling PDF-based query answering with document chunking, vector search, and secure deployment on Streamlit Cloud.

Publications/Presentations

1. Nisarg Trivedi, Ravindrakumar M. Purohit, Satyam Tiwari, and Hemant A. Patil, "CoSSHI: LLM-Powered Large-Scale Multitask Hinglish Dataset for Speech Forensics", Sarawak, Malaysia, 04-06 Aug. 2025. *29th International Conference on Asian Language Processing (IALP). 2025.*
2. Siddharth Kumar, Nisarg Trivedi, Ravindrakumar M. Purohit, and Hemant A. Patil, "BAANI: A 296M-Parameter Neural Vocoder for End-to-End Punjabi Speech Synthesis," *The 17th Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC 2025). 2025.*

3. Selected among the Top 15 teams on the Providing Guidance Track of the BEA Shared Task Dataset at the **Indian Symposium on Machine Learning Datathon held at BITS Pilani, Hyderabad, 2025**
4. Presented a research abstract on "Dynamic Pricing and Multi-Period Inventory Optimization for Perishable Agricultural Commodities" at the **India Management Research Conference, Indian Institute of Management Ahmedabad (IIM-A), 2025**
5. Presented a research poster on "Dyslexia Detection through Handwriting" at the **Conference on Technology for Disabled Persons, Indian Institute of Management Ahmedabad (IIM-A), 2025**

Technical Skills

Languages: Python3, C/C++, HTML/CSS, SQL

Frameworks & Libraries: PyTorch, Scikit-Learn, LangChain, NumPy, Pandas, Matplotlib, Streamlit, HuggingFace

Tools: Git, GitHub, Linux, VS Code, Jupyter Notebook, Overleaf